

Time Complexity Analysis in C

This program demonstrates Constant Time $O(1)$, Linear Time $O(n)$, and Quadratic Time $O(n^2)$ operations. The input size is increased to observe the time consumed by each operation.

C Program

```
#include <stdio.h>
#include <time.h>

int main() {
    int sizes[] = {1000, 5000, 10000, 20000};
    int nSizes = 4;

    printf("Input Size\tO(1) Time\tO(n) Time\tO(n^2) Time\n");

    for (int s = 0; s < nSizes; s++) {
        int n = sizes[s];
        clock_t start, end;
        double constantTime, linearTime, quadraticTime;

        // O(1) - Constant Time
        start = clock();
        int x = 100;
        x = x + 1;
        end = clock();
        constantTime = ((double)(end - start)) / CLOCKS_PER_SEC;

        // O(n) - Linear Time
        start = clock();
        long long sum = 0;
        for (int i = 0; i < n; i++) {
            sum += i;
        }
        end = clock();
        linearTime = ((double)(end - start)) / CLOCKS_PER_SEC;

        // O(n^2) - Quadratic Time
        start = clock();
        long long count = 0;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                count++;
            }
        }
        end = clock();
        quadraticTime = ((double)(end - start)) / CLOCKS_PER_SEC;

        printf("%d\t\t%f\t%f\t%f\n",
            n, constantTime, linearTime, quadraticTime);
    }

    return 0;
}
```

Expected Observation:

1. $O(1)$ remains almost constant regardless of input size.
2. $O(n)$ increases approximately in proportion to n .
3. $O(n^2)$ grows much faster as input size increases.